

choptuik_ac_bc · theorems12 ·
github.com/wild8highlander/choptuik_ac_bc

The Identity Theorem:

$$\tau^*(\delta_C) = 1323/(196 - \pi^2) =$$

τ^*_{T1c} and the Soft Pole 45/8

Two routes, one number · simple pole = exact char(2) root · the linear level stays
finite

Author:	Ishak Khamzatovich Isaev
Series:	Monograph No. 08 of the “Twelve Theorems” cycle
Machine:	sympy_center_o7.py + puiseux_soft_pole.py · SymPy, exact arithmetic
Verdict file:	einstein_direct/results/ q5_center_o7.json + q6_puiseux_soft_pole.json
Status:	PROVED EXACT (three routes to

Abstract

This monograph records two exact facts of the OPEN9 v19 campaign. First, the identity theorem: the clock of the surviving brick, computed through the clock-pair GCD on the brick, through the T1c one-brick closure, and through the chain z-closure of the linear system, is one and the same exact number, $\tau^*(\delta_C) = 27/(4 - \pi^2/49) = 1323/(196 - \pi^2) = \tau^*_{T1c} \approx 7.1079202069$ — two independent symbolic routes, one number; the chain closes the linear z-system exactly, and its closure $P4h_{lin} = t/10 + 16t^3/135$ is a polynomial with no singularity.

Second, the soft-pole theorem: the nonlinear chain representation $P4h_{chain}(t) = t^3(352t^2 - 540)/(135(8t^2 - 45))$ has a simple pole at $\tau = 45/8$ (numerator $6075\sqrt{10} \neq 0$), the growing mode $\lambda^+ = 2$ appears in char exactly at $\tau = 1/2$ and $\tau = 45/8$ — the second point is precisely the chain pole — yet the linear level is finite there ($13\sqrt{10}/40$): the pole is a chain-representation artifact and the branch passes through. Tables of the three routes and three simultaneous descriptions, corollaries, honest caveats and the reproduction protocol are included.

Keywords: identity theorem, $\tau^* = 1323/(196 - \pi^2)$, soft pole $45/8$, char(2), chain closure, Puiseux, SymPy, machine theorem.

Table of Contents

1. Introduction.....	2
2. Notation and Setting.....	2
3. Statement of the Theorem.....	2
4. Sketch of the Machine Proof.....	4
4.1. The exact chain closure (i)–(ii).....	4
4.2. The pole classification (iii)–(v).....	4
5. Numerical Results.....	5
6. Corollaries.....	6
7. Honest Caveats.....	6
8. Reproducibility.....	6
References.....	7

Note: this Table of Contents is generated via field codes. To ensure page number accuracy after editing, please right-click the TOC and select “Update Field”.

1. Introduction

This monograph records two exact facts delivered by the OPEN9 v19 campaign on questions No. 5 and No. 6 of the verification monograph: the identity $\tau^*(\delta_C) = 1323/(196 - \pi^2) = \tau^*_{T1c}$ — two independent symbolic routes to one number — and the soft-pole theorem for $\tau = 45/8$: a simple pole of the nonlinear chain representation that coincides exactly with the $\text{char}(2)$ root while the linear level stays finite. The machines are `sympy_center_o7.py` (question No. 5) and `puiseux_soft_pole.py` (question No. 6).

Both facts are stability statements about the corrected system. The identity shows that the clock of the surviving brick, computed twice — once as the common root of the clock pair on the brick (monograph No. 1) and once through the T1c one-brick closure — is the same number in exact arithmetic; the tower's clock does not depend on the route. The soft-pole theorem shows that the frightening singularity $45/8$ of the nonlinear chain is a representation artifact: the linear level is finite there, the pole is simple, the Laurent principal part is machine-written, and the branch passes through (the T1b soft degeneracy).

2. Notation and Setting

The chain z -system of the truncated tower couples the central amplitudes through the chain forms; its linear part admits the P4h closure and its nonlinear part the chain representation `P4h_chain(t)` with parameter $t = T0h$. The linear closure `P4h_lin` is the polynomial part of the same closure; $\text{char}(2)$ denotes the characteristic polynomial of the linear tower pencil evaluated at the growing-mode candidate $\lambda = 2$. The surviving brick is $\delta_C = \pi/7$ with the additive Berry constant $\kappa = 2 - \pi^2/98$.

Question No. 5 asks whether the truncated tower converges; the machine closes the linear z -system exactly through the chain and records a new identity for the clock of the brick. Question No. 6 asks what the pole $45/8$ is; the machine classifies it as a simple pole of the chain representation, proves the exact coincidence with the $\text{char}(2)$ root at the baseline, and demonstrates symbolically that the divergence is entirely in the nonlinear term.

3. Statement of the Theorem

Theorem 8 (the identity and the soft pole). Let $\delta_C = \pi/7$, $\kappa = 2 - \pi^2/98$. Then:

(i) the chain closes the linear z-system exactly: F1, F2, F4, F5 vanish identically in T0h, and F3 produces the linear closure

$$P4h_{\text{lin}}(t) = t \frac{t}{10} \{ 16t^3 \frac{t}{135} \{ (polynomial, no singularity) \} \} \quad (8.1)$$

(ii) the clock of the surviving brick computed through the clock pair on the brick and through the T1c one-brick closure coincide exactly:

$$\tau_C(t) = 27 \frac{t}{4 - t^2/49} \{ 1323 \frac{t}{196 - t^2} \tau_{T1c}(t) \} \quad (8.2)$$

and the T1c clock factor vanishes at $\tau^*(\delta_C)$ exactly (two routes, one number);

(iii) the nonlinear chain representation

$$P4h_{\text{chain}}(t) = \frac{t^3 (352t^2 - 540)}{135 (8t^2 - 45)} \quad (8.3)$$

has a simple pole at $t = 45/8$ (the numerator at the pole equals $6075\sqrt{10} \neq 0$); the Laurent principal part is machine-written;

(iv) the growing mode $\lambda^+ = 2$ appears in char at $\tau = 1/2$ and $\tau = 45/8$ exactly:

$$\text{char}(\lambda=2, \tau) = \frac{8}{27} (2\tau - 1)(8\tau - 45) \quad (8.4)$$

the second point is precisely the chain pole (proved by exact factorization of the characteristic polynomial);

(v) the pole is soft: the linear level is finite at $\tau = 45/8$ — $P4h_{\text{lin}}(t) = t(16t^2 + 27)/270 \rightarrow 13\sqrt{10}/40$ — and the divergence lives entirely in the nonlinear term; the branch passes through (soft degeneracy, multistart $F \sim 10^{-15}$), so the pole is a chain-representation artifact, not a book.

4. Sketch of the Machine Proof

4.1. The exact chain closure (i)–(ii)

The machine substitutes the chain forms into the linear z-system and simplifies symbolically: four of the five closure residuals vanish identically in T0h, and the fifth reduces to the polynomial (8.1) — note that no denominator survives, hence the linear level

cannot diverge anywhere. The identity (8.2) is then proved twice: the clock-pair route (GCD of the books on the brick, monograph No. 1) and the T1c route (one-brick source form, campaign v17) both produce the factor $(196 - \pi^2)T0h^2 - 1323$ up to a non-vanishing multiplicative constant, which SymPy verifies in exact arithmetic over $\mathbb{Q}(\pi)$. The naive O6 incompatibility $\tau_3 = 9/4$ versus $1/\tau_5 = 9/16$ is reproduced as an overdetermined system (center_o6_nsolve) and recorded as the adiabatic artifact it is.

4.2. The pole classification (iii)–(v)

The chain representation (8.3) is constructed symbolically; the denominator $135(8t^2 - 45)$ has the positive root $t = \sqrt{45/8}$ in t , corresponding to $\tau = 45/8$ in the tower variable; the machine evaluates the numerator at the pole — $6075\sqrt{10} \neq 0$ — proving simplicity, and writes the Laurent principal part. The coincidence (8.4) is proved by exact factorization: $\text{char}(\lambda = 2, \tau)$ factors as $(2\tau - 1)(8\tau - 45)$ times a constant, so the growing mode of the linear tower is born exactly at the pole of the nonlinear chain. Softness is proved by comparing the levels: P4h_lin is polynomial (finite at $45/8$, value $13\sqrt{10}/40$), and the divergent part belongs to the nonlinear correction only. The brick sensitivity is recorded: the chain pole moves $45/8 \rightarrow 5.9233$ on the brick, while the $\text{char}(2)$ root is written for the baseline; the brick relation is stated open.

Table 8.1. Layers of the two verdicts (q5, q6)

Layer	Statement	Status
q5 chain	$F1/F2/F4/F5 = 0$ identically; $F3 \rightarrow \text{P4h_lin}$ polynomial	exact
q5 M3-link	O4 on the chain = $2t^2/3$ = the O3 route ($\kappa = 2$)	exact
q5 identity	$\tau^*(\delta_C) = \tau^*_{T1c} = 1323/(196 - \pi^2)$	exact (two routes)
q6 pole	simple pole at $45/8$; numerator $6075\sqrt{10} \neq 0$; Laurent written	exact
q6 char(2)	$\text{char}(\lambda = 2, \tau) = (8/27)(2\tau - 1)(8\tau - 45)$	exact factorization
q6 softness	linear level finite: $13\sqrt{10}/40$ at the pole	exact

5. Numerical Results

Table 8.2. The two routes to τ^* on the surviving brick

Route	Machine	τ^* (numeric)	Clock factor at τ^*
Clock pair on the brick	brick GCD (No. 1)	7.10792020692263	0 exactly
T1c one-brick closure	clock_closure_t1c	7.10792020692263	0 exactly
Chain z-closure (q5)	sympy_center_o7	7.10792020692263	0 exactly

Table 8.3. The pole 45/8: three simultaneous descriptions (baseline)

Object	At $\tau = 45/8$	Verdict
P4h_chain (nonlinear chain)	simple pole; principal part $6075\sqrt{10}/(135 \cdot 8)$	divergent
P4h_lin (linear closure)	$13\sqrt{10}/40 \approx 1.0277$	finite
char($\lambda = 2, \tau$)	root exactly at 45/8	coincidence exact

The level separation of Table 8.3 is the content of “soft”: everything nonlinear diverges, everything linear does not. On the brick the chain pole moves to 5.9233 (numerically), consistent with the exact family $\tau^*(\kappa) = 27/(2\kappa)$ evaluated at the brick constant 1.8992897510 — the same number the ladder of monograph No. 2 gives for $\delta = \pi/7$; the char(2) root at the brick is recorded as an open relation.

6. Corollaries

Corollary 8.1 (route independence of the clock). Three symbolic routes — the clock pair, the T1c closure, and the chain z-closure — give the same τ^* in exact arithmetic. The clock of the tower is a well-defined algebraic object, not a feature of a particular derivation; this is the strongest stability statement available for the truncated tower and the working justification of the two-point protocol.

Corollary 8.2 (the pole is not physics). Since the linear level is finite at 45/8 and the branch passes through, no book or clock may be attached to the pole; any future “physics of 45/8” would have to come from Puiseux branch structure of the nonlinear representation, which remains explicitly not excluded (honest caveat, v16–v17).

Corollary 8.3 (separation of mechanisms). The level-3 exit rate $|\text{DrV}| = (39.2 \pm 0.2) \cdot A$ of monograph No. 6 and the pole $45/8$ are different objects: an amplitude-linear exit rate versus a representation pole. The q6 verdict records this separation explicitly to prevent conflation in future campaigns.

7. Honest Caveats

The identity (8.2) is exact for the truncated tower in the T1c source form; tower enhancement (the O7 programme) may in principle move τ^* , though the convergence status is honestly not established (the levels move $6.75 \rightarrow 7.1079$, $+5.27\%$). The $\text{char}(2)$ coincidence is proved at the baseline; the brick relation of the $\text{char}(2)$ root is open. Puiseux branch jumps at the soft pole are not excluded: the representation may switch branches when continued through $45/8$, and the Laurent analysis is local. The pole moves with the brick ($45/8 \rightarrow 5.9233$); all statements of (iii)–(v) are baseline statements unless explicitly stated. Finally, the linear closure (8.1) closes the linear z -system; nonlinear closures beyond the chain are the subject of monograph No. 6.

8. Reproducibility

Table 8.4. Reproduction protocol for Theorem 8

Step	Command	Output
1	<code>python3 sympy_center_o7.py</code>	results/q5_center_o7.json (chain closure + identity)
2	<code>python3 puiseux_soft_pole.py</code>	results/q6_puiseux_soft_pole.json (pole classification)
3	<code>python3 clock_closure_t1c.py</code>	results/clock_closure_t1c.json (T1c route)
4	<code>python3 -m pytest tests/ -q</code>	58 passed (q5/q6 tests included)

Both q-machines run in seconds (offline, deterministic, BLAS = 1). The data correspond to commit 5b256cd; self-contained copies live in `einstein_direct/open9/machines/`.

References

- [1] Repository wild8highlander/choptuik_ac_bc: `einstein_direct/sympy_center_o7.py`;
`einstein_direct/results/q5_center_o7.json` (OPEN9 v19, question No. 5).
- [2] Ibid.: `einstein_direct/puiseux_soft_pole.py`; `einstein_direct/results/q6_puiseux_soft_pole.json`
(OPEN9 v19, question No. 6).
- [3] Ibid.: `einstein_direct/clock_closure_t1c.py`; `einstein_direct/results/clock_closure_t1c.json` (T1c
route, campaign v17).
- [4] Ibid.: `einstein_direct/README_EN.md` §§28, 30–31.
- [5] Monographs No. 1–2, No. 6 of the theorems12 cycle.